3.1 Unit description

Introduction	This unit requires that students undertake **either** a case study involving an application of physics and a related practical, **or** a physics-based visit and a related practical. The teacher, not the student, identifies the visit or case study that students will be doing. All candidates may do the same case study or the same visit; however it is vital that candidates demonstrate that the assessed work that they produce is entirely their own work. This unit may be completed at any time during the AS course but it would be more appropriate to administer this assessment near the end of the AS year.
Case study	Edexcel will provide case studies for five different topics. Centres may either use one of the case studies provided by Edexcel or devise their own case study to match local needs and the interests of their candidates. Centre-devised case studies will not require approval from Edexcel; however, it is the responsibility of the centre to ensure that centre-devised case studies match the assessment criteria for this unit and that students have the opportunity to gain all the marks in the mark scheme. Candidates may all do the same case study or they may do different case studies. If all candidates do the same case study then they must ensure that work submitted for assessment is their own. There should be a connection between the case study and the practical work that is undertaken for this unit. For example a case study might be based on an application of Quantum Tunnelling Composite. This would offer the opportunity for practical work relating compressive force to resistance in this type of material. Ideally the case study should deal with concepts covered within the AS specification but this is not a requirement for the assessment of this unit.
Visit	The visit is intended to bring candidates into direct contact with a real-life example of physics in use. There should be a connection between the visit and the practical work that is undertaken for this unit. For example candidates might visit a church or concert hall. A related practical could be to investigate the relationship between the length of an organ pipe (using a glass tube to represent the organ pipe) and the frequency of its sound at resonance. The teacher or the host may provide briefing materials for the visit.